

2026 IEEE SUSTAINABLE POWER AND ENERGY CONFERENCE

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SPECIAL SESSION 4 (SS4) • AMDDC-HES

Advanced Multiport DC-DC Converters for Integrated Hybrid Energy Systems and Carbon Neutrality

— Call for Papers —

Organized and Co-Chaired by:

- **Dr. Sivaprasad Athikkal** — Associate Professor, Dept. of EEE, Muthoot Institute of Technology and Science (Autonomous), Kochi
- **Dr. Gangavarapu Gurukumar** — Research Consultant, GE Vernova, John F. Welch Technology Centre, Bengaluru
- **Dr. Kumaravel Sundaramoorthy** — Professor, Dept. of Electrical Engineering, NIT Calicut

Session Overview:

Multiport DC-DC converters have emerged as high-density, cost-effective solutions for managing multidirectional power flows across diverse renewable sources, storage units, and DC loads through a unified conversion stage — overcoming the excessive component counts and losses of conventional single-input single-output converters. This session addresses circuit topologies, decoupled power-flow control, wide-bandgap device integration, and hardware implementations for renewable microgrids, hybrid electric vehicles, and carbon-neutral energy systems.

Topics of Interest Include, but Are Not Limited to:

- Isolated, non-isolated, and reconfigurable multiport DC-DC topologies
- High-gain, bidirectional multiport converters for PV and fuel-cell integration
- Decoupled power management and multi-loop control strategies
- Soft-switching (ZVS/ZCS) techniques and efficiency optimization in MDCs
- Wide-bandgap (SiC/GaN) power semiconductor applications
- Multiport architectures for hybrid energy storage (battery + supercapacitor)
- Multi-source EV fast-charging and V2G/V2X multiport interfaces
- Energy management algorithms for multiport-based DC microgrids
- Fault ride-through, protection coordination, and reliability of MDCs
- Real-time hardware-in-the-loop (HIL) validation and modeling

Important Dates:

Paper Submission Opens	25 August 2026
Paper Submission Deadline	30 September 2026
Acceptance Notification	15 October 2026
Registration Due	1 November 2026
Conference Dates	4–6 December 2026

Format: Half-day technical special session, targeting ~8–10 peer-reviewed papers. Author guidelines and peer-review standards will follow the regular iSPEC 2026 paper submission process. Accepted and presented papers will be submitted for inclusion in IEEE Xplore, subject to IEEE Xplore's scope and quality requirements.

Submission Portal: <https://ispec2026.org/submission>

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